

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

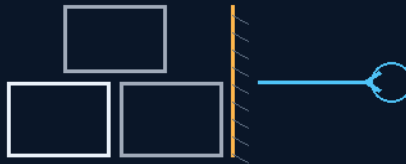
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 274

3D PRINTING-THE FIFTY-YEAR FEEDSTOCK

vaseline and plastic wrap, or vacuum sealed
bandsaw portioning, shredder to the line



first customer: moon regolith sweeper filaments

THE NOT-AIRTIGHT LAW · POWDER COMPOSITE · THE MELT PLANT

ONE SHELF FOR THE WHOLE SHOP

Manufacturing stores — v1.0 in force

GP-IM-274

Revision v1.0 — 24 August 2026

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SEATED LAW — EXTERNAL VET PENDING

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 274

PHASE 274: 3D PRINTING-THE FIFTY-YEAR FEEDSTOCK — BLOCKS, NOT SPOOLS — STATUS: CURRENT — SEATED LAW / EXTERNAL VET PENDING (THE NOT-AIRTIGHT LAW SEATED; THE MELT PLANT RIDES THE 129/130 CENTRIFUGAL HERITAGE). Blocks, not spools: feedstock stored as solid blocks — Vaseline and plastic wrap, or vacuum sealed — good for fifty years, portioned on a bandsaw, shredded to the cord and filament line. Metals ride as powder composite in the same blocks: one storage form for the whole shop. The not-airtight law is the honesty: the wrap is a dust and oxygen barrier, not a hermetic seal — the phase says so and prices the difference. The zero-g melt plant rides the 129/130 centrifugal heritage: the cake-oven g-force gives the melt a down, so the filament line extrudes like it has a floor. The first customer is seated: moon regolith sweeper filaments — the feedstock that lands first is the feedstock the sweepers eat.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-274, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the feedstock law as seated (contents line, the blocks in crayon — the wrap, the bandsaw, the shredder, the powder composite, the melt plant, the not-airtight law, the first customer — entire) — §2 the physics, graded — §3 the system in the stores — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 274: **3D Printing-The Fifty-Year Feedstock — Blocks, Not Spools.** Long-shelf-life 3D-printer feedstock doctrine: store bulk polymer and metal as Vaseline-sealed, wrap- or vacuum-sealed blocks (50-year shelf vs spool degradation and powder factorial oxidation); bandsaw-portion (metal saw for metal), shredder-to-premelt-to-screw-extruder for cord/filament — industrial regrind practice, all current tech; metals as bound powder composite (PIM), gas atomization fastest

powdering; ship-side zero-g adaptations: cake-oven g-force crucibles (Phase 129/130 heritage), gas-injection pressure-vent to rotating-wall-screw collection lines, sleeve field oscillation anti-caking, powder stock sealed but NOT airtight (displacement-airflow feed physics); bound powder optimal for metal parts; high-tensile polymer filament serves the moon regolith sweeper.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

THE LAW: store bulk, cord on demand. The shelf life of a 3D-print program is set by its feedstock, and spooled feedstock dies on the shelf — polymer hydrolyzes and embrittles like old whipper-snippier cord, and metal powder, being all surface, oxidizes factorially faster than bulk. The Protocols' answer is the block: bulk stock, sealed, portioned and powdered only at the point of use. (The author's full dictation and the audit's grade: sitting 179.)

- STORAGE: blocks of polymer or metal, covered in Vaseline and sealed in plastic wrap or vacuum sealed. Fifty-year shelf is the design target; bulk material has no spool geometry to degrade and no powder surface to oxidize, and the grease barrier is century-proven practice.
- PORTIONING: blocks go through a bandsaw for the amount required — metal saw for metal.
- CORD/FILAMENT LINE: portions through the shredder (the plastic factory's regrind practice for faulty items), into the premelt, into the screw-fed extruder — et voila, cord/filament. All current tech; the screw feeder is the standard feeder.
- METALS: treated the same — all 3D-printer metals are powder composite, so industrial-level printing is available. A dozen powdering methods exist; gas (atomization) is the fastest. Bound powder is optimal for metal parts.
- ZERO-G MELT PLANT (ship-side): the cake-oven g-force system for crucible melts (Phase 129/130 centrifugal heritage); gas injection pressure-vented to collection lines of rotating-wall-screw mechanics; sleeve field oscillation on the walls to prevent caking.
- THE NOT-AIRTIGHT LAW: powder stock is sealed for zero-g housekeeping — but NOT airtight. Nothing flows in a sealed unit into a heated, pressurized melt and extrusion zone without displacement air flow; trying is like trying to suck feed out of a vacuum box. Sealed against powder escape, vented for flow.
- FIRST CUSTOMER: simple high-tensile polymer filaments for the moon regolith sweeper — whose lunar print program (Phase 264 family, first-drop manifest) needs exactly the shelf life spools cannot promise and blocks can.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

BLOCKS, NOT SPOOLS: a spool is a storage accident — tangles, voids, moisture traps; a block is a brick of pure feedstock with none of them. The bandsaw portions, the shredder feeds the cord and filament line, and the block sits fifty years between the two.

THE PRESERVATION IS SHOP-GRADE AND HONEST: Vaseline and plastic wrap, or vacuum sealed — the not-airtight law stated in the open: the wrap is a dust and oxygen barrier, not a hermetic seal, and the fifty-year claim is priced against that difference.

METALS RIDE AS POWDER COMPOSITE: one storage form for the whole shop — polymer blocks and metal-powder composite blocks on the same shelf, portioned the same way, fed to their own lines.

THE ZERO-G MELT PLANT RIDES THE HERITAGE: the 129/130 cake-oven g-force gives the melt a down — centrifugal heritage spent on extrusion, so the filament line behaves like it has a floor.

THE FIRST CUSTOMER IS SEATED: moon regolith sweeper filaments — the feedstock that lands first feeds the machines that clean the ground it landed on. The loop closes at the sweepers.

§3 — THE SYSTEM IN THE STORES

- **The form:** blocks, not spools — fifty years on the shelf, no tangles, no traps.
- **The wrap:** Vaseline and plastic wrap, or vacuum sealed — the not-airtight law in the open.
- **The portioning:** bandsaw, then shredder to the cord and filament line.
- **The metals:** powder composite in the same blocks — one shelf for the whole shop.
- **The melt plant:** zero-g, riding the 129/130 centrifugal heritage — the cake-oven g-force as the floor.
- **The first customer:** moon regolith sweeper filaments — the loop closes at the sweepers.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- Phases 129/130 are the heritage: the centrifugal literacy (the cake-oven g-force) giving the melt plant its down — rotation priced honestly, one family up.
- 265's aquaculture screens and fittings are a customer: the block feed prints the plumbing the tank bench needs.
- 258's foundry is the Moon-side sibling: the foundry makes stock from regolith; the feedstock doctrine ships and stores what the foundry cannot.
- The not-airtight law is the file's honesty habit: the difference between wrap and hermetic stated on the page — the fifty-year claim priced against it.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Cast the blocks: polymer and metal-powder composite — the shop's two lines, one shelf.
- 2. Wrap per the law: Vaseline and plastic wrap, or vacuum sealed — the not-airtight difference priced.
- 3. Fit the bandsaw: portioning to the line's feed rate — the block cut like timber.
- 4. Shred to line: the cord and filament line fed from the shred — melt plant on the centrifuge heritage.
- 5. Book the first customer: moon regolith sweeper filaments — the first shipment's manifest already named.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phases 129/130 (the centrifuge heritage):** the down — the melt plant's floor is the cake-oven g-force.
- **Phase 258 (water from rocks):** the Moon-side sibling — foundry stock versus shipped blocks, priced together.
- **Phase 265 (the aquaculture tank):** a customer — screens and fittings from the block feed.
- **The sweeper line (regolith sweepers):** the first customer — the filaments the sweepers eat.

§8 — DO-NOT-BUILD

- Do NOT ship spools — blocks, not spools; the tangle is a storage accident this phase refuses.
- Do NOT call the wrap hermetic — the not-airtight law: dust and oxygen barrier, priced as such.
- Do NOT split the shelf — polymer and powder composite in the same block form; one storage doctrine.
- Do NOT melt without a down — the centrifuge heritage is the floor; a floating melt is a hazard.
- Do NOT let the first shipment land unbooked — the sweeper filaments are the named first customer.

§9 — OWED

OWED: the bench list, whole — block size and wrap spec per material; the fifty-year sample program

(the claim benched, not asserted); bandsaw blade spec per material; the shred-to-line feed rates; the melt plant's centrifuge rate from the 129/130 tables. And the external vet, when it lands.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-274, v1.0 — CURRENT — SEATED LAW / EXTERNAL VET PENDING (THE NOT-AIRTIGHT LAW SEATED; THE MELT PLANT RIDES THE 129/130 CENTRIFUGAL HERITAGE), seated law, master v2.1 (numerical print order). Built 24 August 2026, fifteenth of the 20-chapter 'you get the next 20 ready' shift (the author's order, 24 August 2026, seated in sitting 623 — 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618, 622 and 623): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20' / 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 274